ALGEBRAIZATION OF FORMAL ISOMORPHISMS AWAY FROM FINITE TORSION, NOETHERIAN SPACES, EASY CASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9800-9817 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9819-9825, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Definitional/descent-transfer family; part of the chapter's formal-algebraization suite feeding the spaces proper-over-complete card L16140.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale descent/uniqueness for coherent formal systems on spaces (chapter formal-categories setup); Cohomology of Schemes, lemma existence-easy (the schemes-side algebraization theorem).

Source reference. Original lemma 0889: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9825–9825. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S and let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Let $\mathcal{G}$ be a coherent $\mathcal{O}_X$-module, $(\mathcal{F}_n)$ an object of $Coh(X, \mathcal{I})$, and $\alpha : (\mathcal{F}_n) \to \mathcal{G}^\wedge$ a map whose kernel and cokernel are annihilated by a power of $\mathcal{I}$. Then there exists a unique (up to unique isomorphism) triple $(\mathcal{F}, a, \beta)$ where*

(1) *$\mathcal{F}$ is a coherent $\mathcal{O}_X$-module,*
(2) *$a : \mathcal{F} \to \mathcal{G}$ is an $\mathcal{O}_X$-module map whose kernel and cokernel are annihilated by a power of $\mathcal{I}$,*
(3) *$\beta : (\mathcal{F}_n) \to \mathcal{F}^\wedge$ is an isomorphism, and*
(4) *$\alpha = a^\wedge \circ \beta$.*

Proof. The uniqueness and étale descent for objects of $Coh(X, \mathcal{I})$ and $Coh(\mathcal{O}_X)$ implies it suffices to construct $(\mathcal{F}, a, \beta)$ étale locally on X. Thus we reduce to the case of schemes which is Cohomology of Schemes, Lemma 0889. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.